

INTRODUCTION

NOTE: BEFORE WORKING, CLEANING OR MAINTAINING THE EQUIPMENT, IT IS HIGHLY IMPORTANT TO LOCK OUT ALL OF THE ENERGY SOURCES ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

ATTENTION!

NEVER ADD OIL TO THE SYSTEM BEFORE FIRST ENSURING ALL WATER HAS BEEN DRAINED. SET ALL VALVES ACCORDING TO “SYSTEM DRAINING” AND CHECK FOR WATER BEFORE ATTEMPTING TO FILL THE SYSTEM WITH COOKING OIL. PERSONELL INJURY AND MACHINE DAMAGE COULD OCCUR.

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SAFETY NOTICE

- **DO NOT** remove any panels, covers, or door while machine is running.
- **DO NOT** disconnect, remove, or otherwise disable any of the safety interlocks.
- **DO NOT** probe with **anything** into areas where parts or belts are moving (including interiors of panel enclosures, behind chain/sprocket or belt guards).
- **DO NOT** disconnect, jam, or otherwise defeat the operation and purpose of the safety devices such as:
 - Limit switches
 - Door or panel closure switches
 - Safety interlocks
- **DO NOT** place hands or **anything else** into any part of the machine that is in operation or energized and capable of being turned on by accident; or where process material (s) are mixed, fed or other wise worked or conveyed in some manner.
- **DO NOT** allow spilled oil or grease to remain on the floor. Wipe it up before someone slips on it.
- **DO NOT** lift heavy objects alone. Use a forklift or get assistance.
- **DO NOT** push on wrenches. Always try to pull so that if your hand slips you won't injure your knuckles.
- **DO NOT** stand on anything unstable when working above ground level.
- **DO NOT** trust someone else to disconnect the power source. Always check for yourself.
- **DO** wear eye protection when using power tools and when working in the prone position.
- **DO** use ear protection when necessary.
- **DO** plan your job carefully, correctly assemble all components, and double checking all work performed.
- **DO** know where the nearest fire extinguisher is and how to properly use it. *(It is required to have a Class B Extinguisher).*
- **DO** learn CPR and general emergency procedures. Encourage your fellow employees to learn them also.
- **DO** carry out work in a logical sequence and make sure that everything is correctly assembled.

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INTRODUCTION

Congratulations and thank you on your selection of the Lawrence Equipment **CHIP FRYER**. This machine was designed with versatility and efficiency in mind, and along with stainless steel construction and components rated for extreme duty, means your fryer will be running trouble free for many years to come.

With the fluctuating buying habits of today's consumer, it is no longer a sure thing to anticipate a days production. The production quantities and raw materials will vary from day to day. The versatility of this machine, when properly operated and utilized will allow you the production flexibility needed to meet these verifying conditions. As with all machines dealing in variables, the machine will require some minor adjustments to keep the machine operating to its top performance.

As with any machine that must deal with variables, your **CHIP FRYER** will require some minor adjustments to keep the machine operating to its utmost efficiency. To ensure simplified maintenance and dependability, the machine must be: operated properly, kept clean, be properly adjusted, and maintained as outlined in this manual. Cleaning and sanitizing procedures are equally important because they help prevent premature wear and malfunction.

We would like to help you keep your machine in top condition, but to do so you need to adjust your maintenance schedule to match your production schedule. While some plants may use their equipment upwards of 120 hours per week, others may require far less usage to meet their production goals. Herein lies the need to set preventive maintenance schedules that will best match your production rates. Two thousand hours can come as soon as sixteen weeks or take as long as a year. Because of the different ways people use their machines, maintenance needs vary. Read the Preventive Maintenance Section to determine the proper maintenance interval for your machine. Keep in mind that taking the necessary preventive measures contribute to the durability and proper function of the machine.

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THEORY OF OPERATION

The Lawrence Equipment fryer uses a high efficiency heat exchanger to heat and maintain cooking oil at optimal frying temperature.

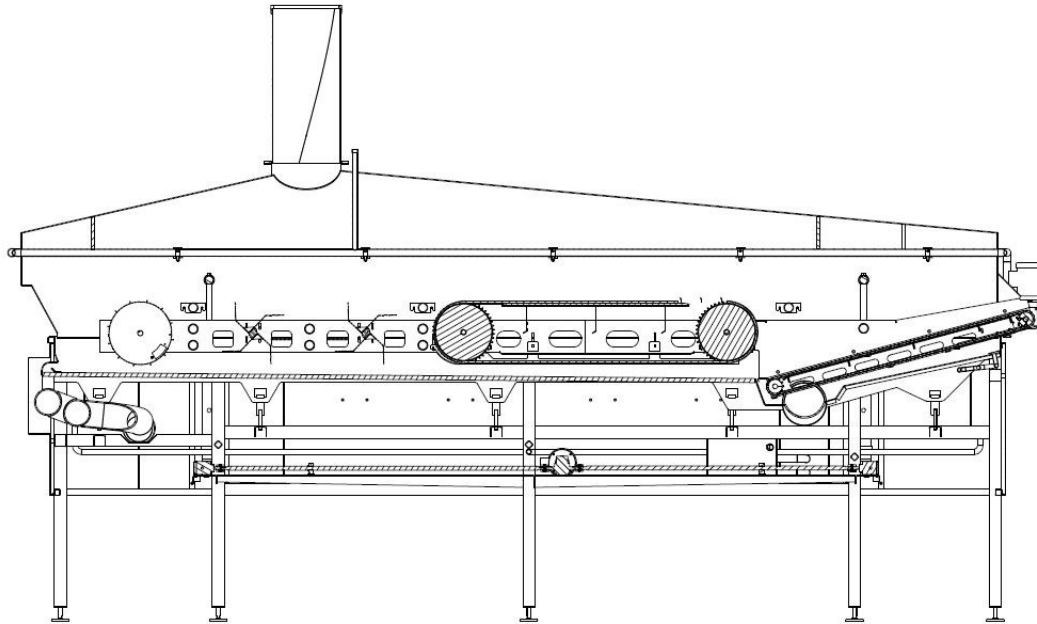
The heated cooking oil is pumped into fryer through a manifold located at the infeed end of the tank. Here the chips are introduced and the force of the oil flow starts them moving forward through the fryer.

Conveyor systems and paddles are used to insure consistent and even frying of the product as well as aiding in product flow.

Once cooked, the product is conveyed out of the cooking oil via a mesh belt. As the product leaves the oil the oil is pumped back out of the fryer and through a continuous centrifugal filtering system. After filtering the oil is re-circulated through the heat exchanger and returns once again to the fryer.

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SPECIFICATIONS

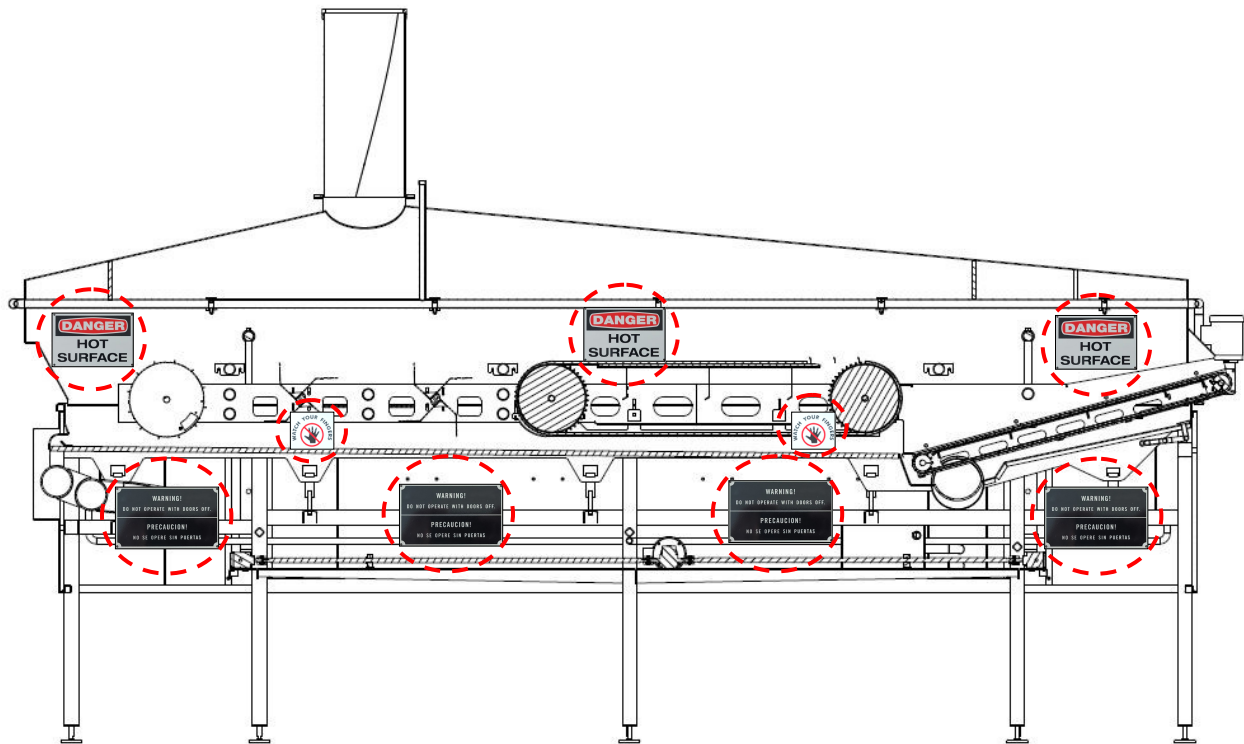


- * The Chip Fryer Serial Plate is located on the Fryer Main Control Panel door. The Serial Plate includes the Model, Serial number and Electrical ratings. It is important to know the Model and Serial numbers of the machine when contacting Lawrence Equipment for parts, service, or general questions.

APPLIANCE TYPE		DESIGN ALTD	
MODEL		SERIAL	DATE
VOLTS		AMPS	
PH		WIRE	
HZ		H.P.	
MAIN GAS PSI		W.C.I.	
B.T.U.			
MIN & MAX MANIFOLD GAS PRESSURE		PURGE TIME	
AIR CFM/P.S.I.		WATER LINE COLD/HOT	
STEAM P.S.I.		LBS/PER HR.	
SHORT CIR. CURRENT RATING			
LAWRENCE EQUIPMENT		2034 NORTH PECK ROAD SO. EL MONTE, CA 91733 (626) 442-2894 (800) 423-4500 Toll Free (except CA) FAX (626) 350-5181	

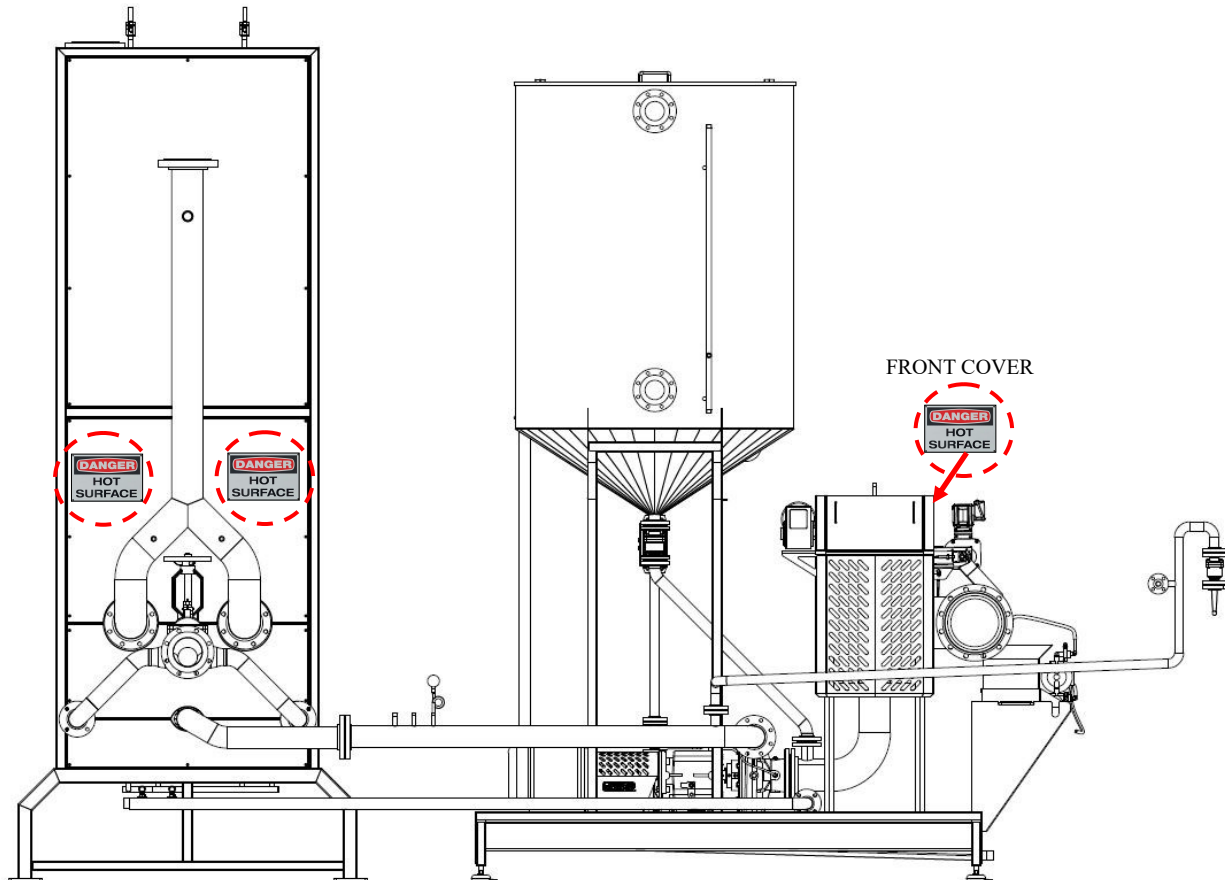
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CHIP FRYER LABEL LOCATIONS



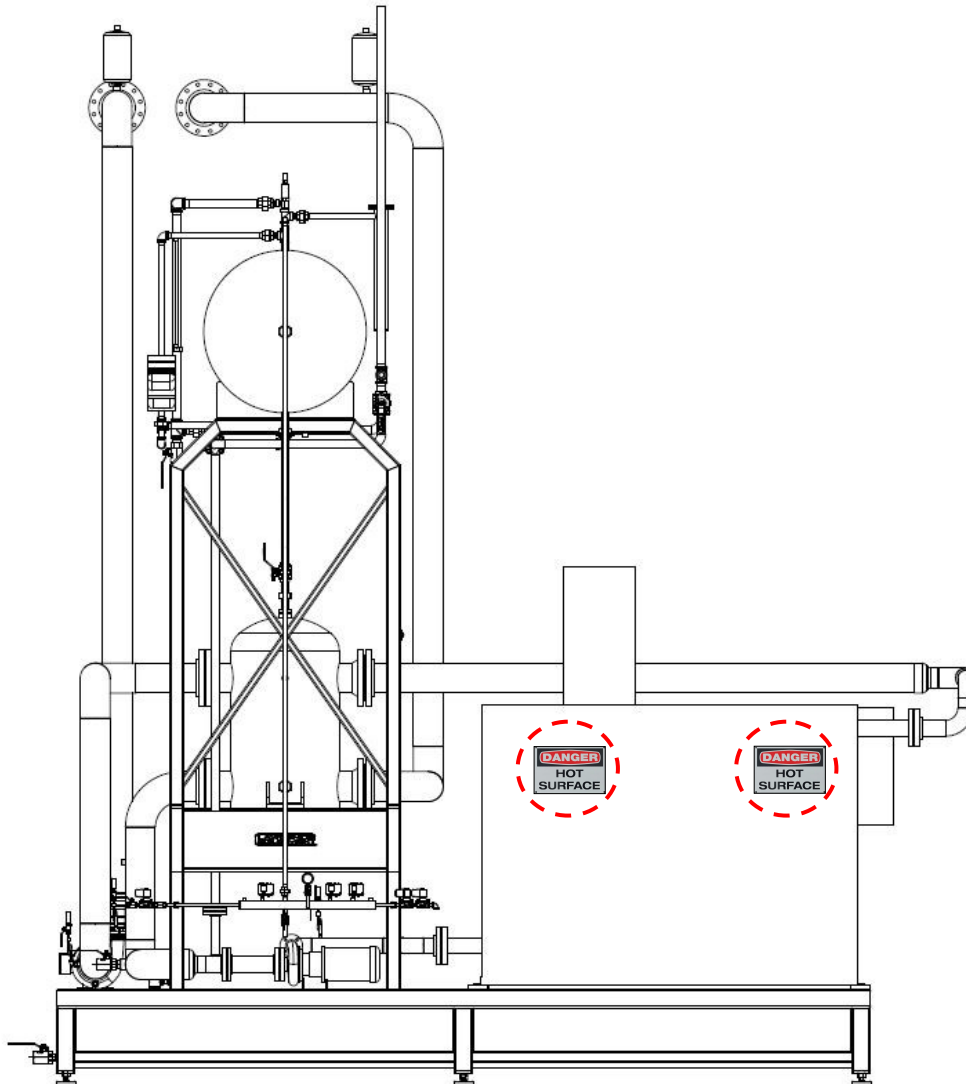
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SUPPORT MODULE LABEL LOCATIONS



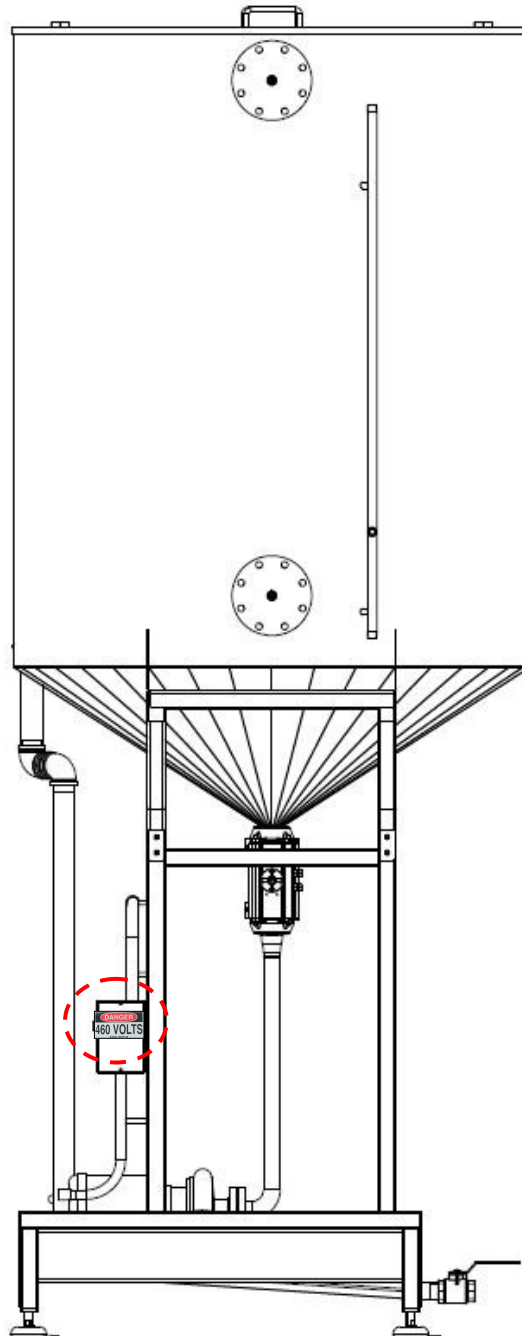
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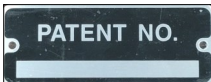
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OIL STORAGE TANK LABEL LOCATIONS



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SAFETY LABEL DESCRIPTIONS/LE PART NUMBERS

Warning Plates/Labels	Description	Serial Plate/Label #
	IMPORTANT! This machine must be properly grounded <u>before</u> operation.	NP001-04
	DANGER! Disconnect all Electrical power before working on or cleaning this machine.	NP001-06
	WARNING! Do not operate with doors off.	NP001-05
(Example) 	Danger 460 Volts High Voltage that could cause serious injury or death.	NP002-40 (460 volts) NP002-74 (400 volts) NP002-67 (380 volts) NP002-38 (115 volts)
	Danger Hazardous Voltage Lock out before servicing	NP003-47
	High Heat conditions and burn risks with hot surface.	NP002-36
	Patent Number	NP001-14
	Potential hazard due to presence of high voltage wires and warns of potential electric shock.	NP003-77 (M) NP003-78 (S)

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

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SAFETY LABEL LOCATIONS

Warning Plates/Labels	Description	Serial Plate/Label #
	Watch your fingers keep and hands and fingers clear of machinery.	NP002-43

Safety Features

- Moving parts, with the exception of all necessary conveying surfaces are covered and guarded.
- Doors are equipped with limit switches for safety. When any door is open, gas burners will automatically shut off. However, DO NOT open doors while running the machine. The slat belt conveyor continues to run at a high temperature.
- There are emergency stop buttons available. They are usually found in control panels or in remote locations throughout the machines.

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BOILER 1 PUMP/MOTOR ROTATION DIRECTION



Warning Plates/Labels	Description	Serial Plate/Label #
	Motor Pump #1 Arrow Indicates Rotation Direction (Clockwise)	N/A

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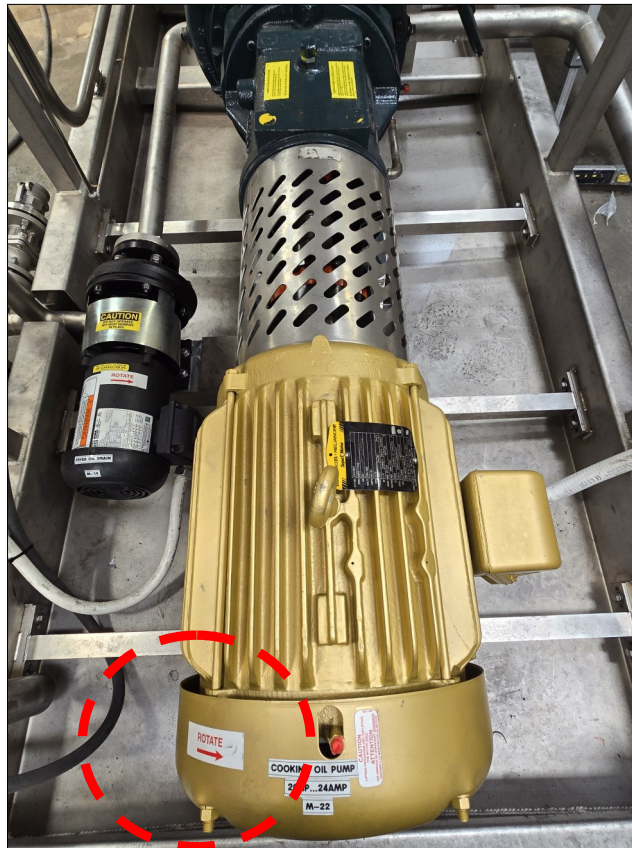
BOILER 2 PUMP/MOTOR ROTATION DIRECTION



Warning Plates/Labels	Description	Serial Plate/Label #
	Motor Pump #2 Arrow Indicates Rotation Direction (Clockwise)	N/A

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COOKING OIL PUMP/MOTOR ROTATION DIRECTION



Warning Plates/Labels	Description	Serial Plate/Label #
	Cooking Oil Pump Fan Direction Arrow Indicates Direction (Clockwise)	N/A

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THERMOFLUID PUMP/MOTOR ROTATION DIRECTION



Warning Plates/Labels	Description	Serial Plate/Label #
	Thermo-fluid Pump Fan Direction Arrow Indicates Direction (Counter Clockwise)	N/A

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FRYER OIL DRAIN PUMP/MOTOR ROTATION DIRECTION



Warning Plates/Labels	Description	Serial Plate/Label #
	Fryer Oil Pump Fan Direction Arrow Indicates Rotation (Clockwise)	N/A

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INSTALLATION

1. **INSPECTION-** Check the equipment for damage during shipping. If you see any damage, note it on the bill of lading before you allow the driver to leave; also notify Lawrence Equipment. Check shipping list with number of crate packages and parts to ensure all items have arrived. Also check for hardware such as nuts, bolts, and electrical connections. For missing parts, report them to Lawrence Equipment immediately.
2. **LOCATION AND INSTALLATION-** Install the machine on a hard floor with sufficient load bearing capacity; it should also satisfy the following conditions:
 - a. Smooth and leveled floor on a solid foundation
 - b. Adequate ventilation.
 - c. Sufficient space should be accessible for machine operation and maintenance
 - d. Install according to layout if prepared by Lawrence Equipment.

NOTE: IF ANY CHANGES ARE MADE DURING INSTALLATION, CONSULT THE FACTORY.

3. **MACHINE ASSEMBLY-** Complete any final assembly such as: installing pipe legs, leveling screws, casters, and other mechanical adjustments required.
4. **HOOD VENT STACK-** Install the hood stack, or type I hood if required by city code. Be sure the correct stack is used. If possible, do not install with excessive bends. A straight stack is preferred, as it reduces flow losses. Allowances must be made for raising the hood. To ease cleaning and maintenance the fryer hood is able to be raised using the hood lift system. Total travel is just under 24" and must be accommodated for during installation. Refer to suggested stack drawing as a guide.

Note: The suggested stack drawing is provided for reference only, all local city and state codes must be satisfied by responsible contractor during installation.)

5. **ELECTRICAL INSTALLATION** - Connect the electrical supply power using the proper wire size and voltage according to applicable code requirements and needs of machine. DO NOT change unit wiring. The wiring should be done by properly trained and licensed personnel.

NOTE: DO NOT TOUCH STACK/HOOD WHILE UNIT IS OPERATING, TEMP, OF HOOD/STACK MAY EXCEED 600 DEGREES FAHRENHEIT.